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title=Software Engineering 2
Requirements Analysis and
Specification Document, name1=Riccardo Piantoni - 10816545, name2=Matteo Rossi - 10798975, name3=Jacopo Sacra

Introduction
Purpose The transition from academia to the job market often presents significant challenges for university students, as
Students&Companies (S&C) is designed to fill this gap. The platform aims to facilitate entrance into the job market
Goals
In this section, there are defined the goals that the system has to achieve:
Profile Management
[**G1**] Students, companies and universities should manage their profiles on the platform.
Internships Publication and Search
[**G2**] Companies should advertise their internships.
[**G3**] Students should apply for internships.
Recommendations
[**G4**] Students should receive recommendations about internship offers matching their CVs.
[**G5**] Companies should receive recommendations about students matching their preferences.
Selection Process
[**G6**] Companies should manage the selection process aimed at evaluating candidates for their internship offers.
Monitoring
[**G7**] Both students and companies should monitor the evolution of the internships they are taking part in.
[**G8**] Both students and companies should report complaints about the ongoing internships.
[**G9**] Universities should handle complaints about the internships of their students.
[**G10**] Both students and companies should provide feedback regarding the concluded internship they have taken part in
Submission Suggestions
[**G11**] Students should receive suggestions about how to make their CVs more appealing.
[**G12**] Companies should receive suggestions about how to make their internship offers' descriptions more appealing.
Scope
Students&Companies (S&C) is a platform that acts as an intermediary system facilitating the internship matching p
The platform defines **Recommendation** as the automated process of identifying suitable internship opportunities for s
The system automates key activities such as generating recommendations using various mechanisms, coordinating interv
World Phenomena
[**WP1**] Students redact their CV.
[**WP2**] Companies make new internship positions available.
[**WP3**] Students decide to take an internship.
[**WP4**] Companies select candidates to be interviewed among the applicants for each of their open positions.
[**WP5**] Companies select the student candidates who fit the most according to the results of their interviews.
[**WP6**] Students carry on their internships at their companies.
[**WP7**] A problem arises in an ongoing internship.
[**WP8**] Universities handle complaints (eventually interrupting internships).
Shared Phenomena
World Controlled
[**SP1**] Students, companies and universities sign up to the platform.
[**SP2**] Students, companies and universities log into the platform.
[**SP3**] Users provide information about themselves to the system.
[**SP4**] Companies provide information about their internship positions to the system.
[**SP5**] Students submit filters to search for suitable internship positions.
[**SP6**] Students proactively apply for an open internship position.
[**SP7**] Students track received recommendations about internship offers.
[**SP8**] Companies track recommendations about potential candidates for their internship offers.
[**SP9**] Students accept recommendations for internships.
[**SP10**] Students reject recommendations for internships.
[**SP11**] Companies accept recommended candidates for their internships.
[**SP12**] Companies reject recommended candidates for their internships.
[**SP13**] Companies contact selected students to set up an interview with them.
[**SP14**] Companies submit questions to students.
[**SP15**] Students provide the information required by companies during the interview.
[**SP16**] Companies finalize the selection process.
[**SP17**] Students track the outcomes of the interviews they have participated in.
[**SP18**] Students and companies report information about an ongoing internship.
[**SP19**] Students, companies and universities monitor an ongoing internship.
[**SP20**] Students and companies report complaints and problems about ongoing internships.
[**SP21**] Universities report information about the problems they have handled.
[**SP22**] Students and companies provide feedback and suggestions about internships.
[**SP23**] Students ask for suggestions about their profiles.
[**SP24**] Companies ask for suggestions about their internship offers.
Machine Controlled
[**SP25**] The system shows to the students the internship offers which match their selection criteria.
[**SP26**] The system shows to the companies the list of all the students who have applied for their internship offers.
[**SP27**] The system shows to students and companies the list of all their received recommendations along with their sta
[**SP28**] The system forwards communications about the scheduling of interviews from companies to students (and vice-
[**SP29**] The system forwards the submitted questions from companies to students.
[**SP30**] The system forwards the submitted answers from students to companies, which collect them.
[**SP31**] The system shows to the candidate students the outcome of their interviews after the conclusion of a selection p
[**SP32**] The system forwards information about the ongoing internships to students and companies.
[**SP33**] The system forwards to universities information about problems in the ongoing internships of their students.
[**SP34**] The system asks for feedback after the conclusion of an internship to improve its recommendation algorithms.
[**SP35**] The system provides suggestions to students about their profiles.
[**SP36**] The system provides suggestions to companies about the description of their internship offers.
Definitions, Acronyms, Abbreviations
System, Platform: these terms are used interchangeably when referring to the system-to-be-developed.

The platform's user interface shall be designed to ensure usability and interactivity, adhering to web standards, briefly d
Below is a description of the primary interfaces needed, as visually displayed in the above map, and how they should be
Home Page This page serves as the entry point to the platform and offers:
Sign-Up (Register): A page to start the personalized registration for the three distinct user types.
Sign-In (Login): A page to perform login that redirects authenticated users to their respective dashboards.
User Dashboards Each user type is provided with a personalized dashboard, divided into sections offering different funct
Student Dashboard
Profile: a page displaying all the student's information and including:
Update Profile: an interactive element for updating personal and academic information.
Improve Profile: an interactive element suggesting optimizations based on the content of the profile and CV.
Search Internship Offers: a section that includes a search bar with filtering options to search offers. Each result of th
Manage Internship Recommendations: a section listing the received internship recommendations. Each recommen
Manage Interviews: a page displaying interview invitations. Each invitation is an interactive element, and Students c
Company Dashboard
Profile: a page displaying all the company's information and including:
Update Profile: an interactive element for updating company information.
Publish an Internship Offer: an interactive element to publish new internship offers.
View Internship Offers: a section listing all published internship offers. Each internship offer is an interactive elemen
Update Internship Offer: an interactive element for updating offer information.
Improve Offer: an interactive element to receive optimizations based on the offer details.
Manage Interviews: a page allowing companies to handle interview invitations and post questions to candidates.
Manage Student Recommendations: a section listing student's recommendations. Each recommendation is an inter
Common Functionalities in Student and Company Dashboards
View Ongoing Internships: a section displaying all the ongoing internships. Each internship is an interactive elemen
Report Problems: an interactive element to report problems to be handled by the university.
Provide Information: an interactive element to write relevant information during an internship.
Report Feedback after an Internship: a section to give feedback upon completion of an internship.
University Dashboard
Profile: a page displaying all university-related information and including:
Update Profile: an interactive element for updating information.
Monitor Student Problems: a section displaying issues reported by students during internships. Each report is an in
Software Interfaces
The system is meant to be a platform-independent WebApp, which does not rely on external APIs or essential third-par
It shall function across the most widely adopted operating systems, including but not limited to Microsoft Windows, Ma
It shall function across the most widely used browsers, including, but not limited to, Google Chrome, Opera, Mozilla Fi
It shall interact appropriately with the chosen services or network protocols for email management, in order to carry on
No additional software installations or configurations are required on the user's device beyond the availability of a suppo
Communication Interfaces
The system shall support standard communication protocols for reliable interaction between the components. Specificall
The system shall operate over standard internet connections, including wired and wireless networks (e.g. Ethernet, Wi-F
The system shall support encrypted communication/secure sessions (e.g. through TLS/SSL) to protect sensitive data, in
The system shall utilize a widely adopted web communication protocol (such as HTTPS) to ensure secure and reliable c
The system shall not require users to install additional software to facilitate communication.
The system shall minimize communication latency to provide a responsive user experience, with server response times a
Hardware Interfaces
The system shall ensure compatibility with common user devices and hosting infrastructure, designed to provide perfor
User Devices
The platform shall support access from a wide range of devices commonly used by end-users, including but not limited t
The platform shall ensure responsiveness and usability across different device types and display resolutions, requiring ha
Hosting Infrastructure
The system shall be deployed on an infrastructure capable of supporting:
Simultaneous access by multiple users with minimal latency
Data processing and storage necessary for handling user interactions and background operations
The hosting environment shall be scalable to accommodate growth in usage and computational demands while maintain
Functional Requirements
[H]

Use Cases and Activity Diagrams
*UC1. Sign Up by a Student

The email address is already linked to an existing account: an error message is shown, and the Student is redirected to
The password does not meet the platform's security requirements: an error message is displayed, and the Student is pro
Some mandatory fields are missing: the system doesn't allow the Student to complete the procedure until all the manda
The confirmation link sent to the indicated mailbox expires: all the information previously inserted into the system by t
[H]

*UC2. Sign Up by a Company

The email address is already linked to an existing account: an error message is shown, and the Company is redirected to
The password does not meet the platform security requirements: An error message is displayed, and the Company is pro
Some mandatory fields are missing: the system does not allow the Company to complete the procedure until all the man
The confirmation link sent to the indicated mailbox expires: all information previously inserted into the system by the C
[H]

*UC3. Sign Up by a University

The email address is already linked to an existing account: an error message is shown, and the University is redirected t
The password does not meet the platform security requirements: An error message is displayed, and the University is re
Some mandatory fields are missing: the system does not allow the University to complete the procedure until all the ma
The confirmation link sent to the indicated mailbox expires: all information previously inserted into the system by the U
[H]

*UC4. Log In by a User

The inserted credentials are incorrect: the system displays an error message indicating that the credentials are invalid a
The account hasn't been verified yet: if the User has not confirmed their email and the confirmation link has not expire
[H]

*UC5. Update User Profile

Some mandatory fields are left empty: the system doesn't allow the User to complete the procedure until all the manda
[H]

*UC6. Publish an Internship Offer

Application domain
Tasks to be performed
Required skills
Internship duration
Compensation terms
Location (on-site, hybrid, or remote)
Application deadline

Some mandatory fields are missing: the system doesn't allow the Company to complete the procedure until all the mand
Some information is not compliant with platform guidelines: the system notifies the Company about the issue and requi
[H]

*UC7. Update Internship Offer

Some mandatory fields are missing: the system doesn't allow the Company to complete the procedure until all the manc
Some information is not compliant with platform guidelines: the system notifies the Company about the issue and requi
[H]

*UC8. Search Internship Offers

Domain of interest
Location (on-site, hybrid, or remote)
Internship duration
Compensation terms
Keywords (an offer matches a keyword if it contains that keyword or a semantically similar one in any of its fields).

The system doesn't find any matching result: if none of the active internship offers matches the selected criteria, the sys
[H]

*UC9. Apply to an Internship Offer

The deadline for applying to the internship has expired: the "Apply" button cannot be clicked and the Student has to g
[H]

*UC10. Generate Recommendations

The system doesn't find any match: it terminates the process silently.
[H]

*UC11. Manage Internship Recommendations

[H]

*UC12. Manage Students Recommendations

[H]

*UC13. Manage an Interview

[H]

*UC14. Provide Information for an ongoing Internship

The Party closes the page without completing the process: the system doesn't record any information and displays the
[H]

*UC15. Monitor an ongoing Internship

[H]

*UC16. Report Problems during an Internship

Nature of the problem.
Relevant details about when and how the issue occurred.
Optionally, attachments as images or documents to further describe the problem.

Some mandatory fields are missing: the system doesn't allow the Party to complete the procedure until all the mandato
[H]

*UC17. Handle Problems during an Internship

[H]

*UC18. Report Feedback after an Internship

The Party closes the survey without completing it: the system doesn't record any information and displays the Party's
[H]

*UC19. Suggest Optimizations for a Student Profile

Academic background
Skills
Uploaded CV
Certifications and extracurricular activities

Add additional skills or certifications.
Update academic details or achievements.
Include or enhance descriptions of projects.
Enhance the overall style by improving clarity or content.

No optimizations can be found: the system doesn't provide any suggestions and terminates silently.
[H]

*UC20. Suggest Optimizations for an Internship Offer

Description of the internship
Application domain
Required skills

Refine the description to better specify tasks and requirements.
Improve the application domain by incorporating additional information or further elaborating on the details already pr
Introduce or modify the set of required skills.
Enhance the overall style by improving clarity or content.

No optimizations can be found: the system doesn't provide any suggestions and terminates silently.
[H]

Mapping On Requirements
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Row ID Goal ID Req ID DA ID UC ID

1 G1 R1, R2, R17 DA1 to DA6 UC1

2 G1 R1, R2 DA1 to DA6 UC2

3 G1 R1, R2 DA1 to DA6 UC3

4 G1 to G12 R2 DA1, DA7 UC4

5 G1 R3, R18 DA1 to DA6 UC5

6 G2 R4, R5, R16 DA1, DA7 UC6

7 G2 R2, R6, R7, R14, R15, R20, R21 DA1, DA7 UC7

8 G3 R2, R8, R9 DA1, DA8 UC8

9 G3 R2, R10, R11, R12, R13 DA1, DA8 UC9

10 G4, G5 R2, R16, R17, R18, R19, R20, R24, R38 DA1 UC10

11 G4 R2, R10, R11, R12, R13, R22, R23 DA1 UC11

12 G5 R2, R16, R22, R24 DA1 UC12

13 G6 R2, R25, R26, R27, R28, R29, R30, R31 DA1, DA9, DA10 UC13

14 G7 R2, R32 DA1, DA11 UC14

15 G7 R2, R33 DA1, DA11 UC15

16 G8 R2, R34, R35 DA1, DA12 UC16

17 G9 R2, R36 DA1, DA13 UC17

18 G10 R2, R37, R38 DA1, DA14 UC18

19 G11 R2, R39 DA1 UC19

20 G12 R2, R40 DA1 UC20

Performance Requirements

As the system does not supply any particular critical functionality to its users, it is acceptable if there are relatively sma
Anytime, the system shall be able to handle a load of up to 10000 concurrent Users without significant performance deg
When generating recommendations, both accuracy and F1 score of the mechanisms employed for identifying those recon
Whenever receiving a request, the system shall respond to the requesting User in less than 5 seconds. This doesn't inclu
Whenever any kind of information (including new recommendations, interview invites, or complaints) needs to be availa

Design Constraints

Standards Compliance

This system is designed to adhere to a range of standards and regulations to ensure compliance with legal, technical, an
General Data Protection Regulation (GDPR) The platform ensures compliance with GDPR to protect users' per
Data minimization and anonymization techniques for data storage.
Mechanisms for users to access, modify, and delete their data.
Transparent consent collection processes and data usage policies, ensuring that the user is aware on which operations ar
By integrating these security measures, the S&C platform aims to foster user trust and compliance with current regulat
World Wide Web Consortium (W3C) The system adheres to W3C standards to ensure interoperability and usabil
Use meaningful HTML elements to structure content properly, enabling assistive technologies to interpret the page effec
Follow WAI-ARIA (Web Accessibility Initiative–Accessible Rich Internet Applications) standard, a framework for addin
Perform decisions in order to reduce load times and enhance responsiveness.
Web Content Accessibility Guidelines (WCAG) To ensure accessibility, the system follows the latest WCAG star
Support for assistive technologies: screen readers and magnifiers, speech recognition software, keyboard-only navigation.
Accessible navigation structures and compatibility across various devices and input methods.
Hardware Limitations Considering the nature of the system, which is a WebApp, hardware limitations are minimal, as t

Software System Attributes

The main attributes that the system to be developed shall present are outlined here, listed in order of importance for its

Reliability

The system shall ensure high reliability: namely, it shall handle failures effectively and promptly recover from disruption

Availability

The system shall guarantee at least a 2-nines availability (99%, or a total downtime of at most 3.65 days/year), ensuring

Security

The system shall implement secure authentication and authorization mechanisms to verify user identities and restrict ac

Maintainability

System maintenance shall be planned for hardware or software upgrades, or for the deployment of production code cont

Portability

Since the platform is offered as a WebApp, it operates through standard web browsers and inherently supports compati

Formal Analysis using Alloy Objectives

The formal modeling activity is carried out using Alloy 6 and aims to validate and ensure the soundness of some of the

Consistent System Evolution

We have modeled the rules that define the system's dynamic evolution through time; then, we have defined the intended
Sign Up of a generic User (which can be any of the expected actors: Student, Company or University)